

## Faulted 3 $\phi$ Power Systems

Short circuits in power systems are called **faults**, and are the most interesting and most important type of unbalance and asymmetry in 3 $\phi$  power systems.

Short circuit **faults** in 3 $\phi$  power systems are the most important because they produce the **most severe conditions that can possibly occur** in a power system and can cause catastrophic damage to the power system components.

There are four possible types of short circuit faults:

1. Three Phase (3 $\phi$ ) Faults
2. Line to Ground (L-G) Faults
3. Line to Line (L-L) Faults
4. Double Line to Ground (L-L-G) Faults

When a 3 $\phi$  system has a short circuit fault (or also an open circuit fault), it can usually be very easily ascertained by closely looking at the voltage and current phasors at any port in the system.

Voltage and current phasors at ports closest to the fault location, will more clearly show the fault condition compared to ports far away from the fault.

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**Example:** A fault study was done for large 3 $\phi$  power system using the method of symmetrical components in order to determine how the system electrically reacts to different fault conditions. In one particular analysis, the symmetrical components of the L-G voltages (neutrals are grounded in this system) on the sending end of a transmission line were calculated and are given below.

$$\vec{V}_0 = 646.785 \angle 179.01^\circ V, \quad \vec{V}_1 = 4833.639 \angle -2.40^\circ V, \quad \vec{V}_2 = 4091.573 \angle 177.32^\circ V$$

Are there any faults in this system, and if so then what type(s) of faults?

$$\begin{cases} \vec{V}_{AN} = \vec{V}_0 + \vec{V}_1 + \vec{V}_2 = 95.613 \angle 0.05^\circ V \\ \vec{V}_{BN} = \vec{V}_0 + \vec{a}^2 \vec{V}_1 + \vec{a} \vec{V}_2 = 7824.085 \angle -100.00^\circ V \\ \vec{V}_{CN} = \vec{V}_0 + \vec{a} \vec{V}_1 + \vec{a}^2 \vec{V}_2 = 7768.182 \angle 95.00^\circ V \end{cases}$$

A-phase to neutral voltage is very small compared to the other two line to neutral voltages.

Since neutrals are grounded in this system, the low voltage on A-phase to neutral indicates there is an **A-G fault** in the system.

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**Example:** A 115KV ideal 3 $\phi$  source operates with balanced and symmetric voltage outputs and supplies all of the power to a system. The ideal source is Y-connected with a solidly grounded neutral. An analysis was done of the system using the method of symmetrical components, and the symmetrical components of the source line currents are given below.

$$\vec{I}_0 = 0 \angle 0^\circ A, \quad \vec{I}_1 = 1039 \angle 11^\circ A, \quad \vec{I}_2 = 960 \angle -46^\circ A$$

Are there any faults in this system, and if so then what type(s) of faults?

$$\begin{cases} \vec{I}_A = \vec{I}_0 + \vec{I}_1 + \vec{I}_2 = 1757.16 \angle -16.27^\circ A & \vec{I}_N = 3\vec{I}_0 = 0 \angle 0^\circ A \\ \vec{I}_B = \vec{I}_0 + \vec{a}^2 \vec{I}_1 + \vec{a} \vec{I}_2 = 94.74 \angle -141.03^\circ A \\ \vec{I}_C = \vec{I}_0 + \vec{a} \vec{I}_1 + \vec{a}^2 \vec{I}_2 = 1704.927 \angle 161.11^\circ A \end{cases}$$

Since the source neutral is grounded and there is no neutral current, therefore there cannot be any ground faults (L-G or L-L-G) in the system.

A-phase and C-phase source line currents are very large compared to the source B-phase line current.

Therefore it can be concluded from this data that there is very clearly an **A-C fault** at some location in the system that this source is supplying.